

Genomic Clustering and Survival Analysis in Cancer: A Data-Driven Approach to

Identifying Metastatic Risk Subtypes

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Introduction

Metastasis remains the leading cause of cancer mortality, yet outcomes vary widely even within cancer type¹. Traditional subtyping often fails to capture non-linear interactions between multi-omic features and longitudinal survival²⁻⁴. Building upon prior work that utilized isolated genomic and survival variables, the current framework employs a framework that uses a supervised survival model to learn a patient distance metric based on genomic features and cancer types, then clusters on that representation to identify metastatic-risk subtypes. The objective of this study is to identify genomic subtypes associated with metastatic risk and evaluate their prognostic relevance using survival analysis.

Dataset and Data Cleaning

This study utilized the **MSK-CHORD clinicogenomic dataset**, which contains genomic and clinical information from over 25,000 tumor samples. To define a cohort suitable for metastasis risk analysis, patients were excluded if they (1) had documented metastatic progression prior to or at sequencing, (2) had Stage IV or distant metastatic disease at baseline, (3) had progression recorded at or before sequencing, (4) had no post-sequencing follow-up time, or (5) had incomplete genomic or outcome data. Eligible patients were identified by integrating multiple clinical datasets, including *data_clinical_patient.txt*, *data_timeline_diagnosis.txt*, and *data_timeline_progression.txt*. Genomic features are retained only from primary tumor samples and eligible patients and aggregated at the patient level. Gene-level mutation indicators were derived from *data_mutations.txt*, tumor mutational burden (TMB) and Cancer type from *data_clinical_sample.txt*, and global copy number alteration (CNA) burden was calculated from the segmentation file *data_cnahg19.seg*. Clinical timeline data from *data_timeline_progression.txt* were used to construct a time-to-metastasis (TTM) outcome measured from the date of genomic

sequencing (time 0). Eligible patients were followed from sequencing until first documented metastasis(event) or last recorded follow-up.

Statistical Methods

The primary endpoint of this study was metastasis-free survival (MFS), defined as the time from genomic sequencing to the first documented metastatic event or the date of last follow-up (censoring). The survival outcome for each individual i was represented as a pair (T_i, δ_i) , where T_i denotes the observed follow-up time and δ_i is an indicator variable such that:

$$\text{Surv}(T_i, \delta_i) = \begin{cases} T_i & \text{if } \delta_i = 1 \text{ (metastatic event)} \\ T_i & \text{if } \delta_i = 0 \text{ (censored)} \end{cases}$$

To model the complex, non-linear relationships between these features and MFS, **Random Survival Forest (RSF)** was applied. Predictors included gene-level mutations, tumor mutational burden, copy number alteration burden, and cancer type. The RSF model was implemented using the randomForestSRC package in R, allowing stable estimation of survival relationships within the data.

$$P(T > t \mid T > 0, X)$$

Where:

- T : Time to metastasis.
- X : A feature vector representing the patient's profile, such that $X = \{G, C\}$, where:
 - G = Genomic features (mutations, CNA burden, TMB).
 - C = Clinical covariates (Primary cancer type).
- $T > 0$: The condition that the patient was metastasis-free at the time of genomic sequencing (entry into the cohort).

The RSF model with 1,000 trees was trained, where each tree was built using a bootstrap sample of patients to improve model stability and robustness. For every tree, approximately one-third of patients were excluded from the bootstrap sample and served as out-of-bag (OOB) observations, which were used to obtain unbiased error estimates and to compute patient similarity. A patient-level proximity matrix was then constructed by measuring how frequently pairs of patients appeared in the same terminal nodes across trees, with higher proximity indicating more similar

survival-associated genomic patterns. This proximity matrix was converted to a distance matrix (1 – proximity) and used for clustering by partitioning around medoids (PAM). The optimal cluster number (K=5) was selected via silhouette analysis and validated by bootstrapping (mean Jaccard index = 0.918).

The predictive performance of the Random Survival Forest (RSF) model was evaluated using the concordance index (C-index) and Brier score to assess discrimination and calibration over time in the presence of censored data. To determine the clinical relevance of the identified genomic subtypes, Kaplan–Meier survival curves were generated, with differences in metastasis-free survival distributions assessed via the log-rank test. A multivariate Cox proportional hazards model was further employed to evaluate the independent prognostic value of the clusters while controlling for key clinical covariates, including age, gender, and cancer type. To characterize the biological composition of the subgroups, the association between molecular clusters and primary tumor sites was quantified using Pearson residuals. Finally, cluster-specific RSF analyses were conducted, utilizing Variable Importance (VIMP) scores to identify the specific genomic features driving metastatic risk within individual cancer types.

Results

RSF Model Performance and Cluster Stability: The Random Survival Forest (RSF) demonstrated high predictive accuracy, achieving a Concordance Index (C-index) of **0.769**, indicating strong discriminative power in patient risk-stratification. Over a 3,361-day follow-up horizon, the model significantly outperformed the baseline, yielding an Integrated Brier Score (IBS) of **0.152** (Vs 0.218 for reference). This substantial reduction in prediction error confirms that the RSF framework provides both robust calibration and superior accuracy in estimating survival probabilities over time (Figure 1).

Metastasis-Free Survival and Cluster Identification: Using the RSF-derived proximity matrix, Partitioning Around Medoids (PAM) identified five distinct patient clusters (Figure 2). Log-rank testing confirmed highly significant differences in metastasis-free survival (MFS) across these

cohorts ($p < 0.0001$). Cluster 2 (High-Risk) exhibited the most aggressive trajectory, with a 75% probability of metastasis within 500 days. This phenotype was characterized by a high frequency of TP53 mutations, ERBB2 alterations, and elevated global Copy Number Alteration (CNA) burden. Cluster 4 demonstrated the most favorable long-term MFS, characterized by genomic stability and low genomic chaos, suggesting potential for therapeutic de-escalation

Multivariate Cox Proportional Hazards Analysis: With Cluster 1 as the reference, all subgroups showed a significantly increased hazard of metastasis. Cluster 2 maintained the highest risk profile (HR = 2.22, $p < 2e-16$), representing a 122% increased risk independent of age and cancer type (Figure 3 - Left Panel). The overall model demonstrated robust performance (C-index = 0.679; Global $p < 2e-16$).

Tissue-Specific Enrichment and Biological Composition: Pearson residual analysis revealed significant associations between molecular clusters and primary tumor sites ($p < 0.0001$) (Figure 3-Right Panel). Cluster 2 was heavily enriched for Breast Cancer, correlating with its aggressive clinical course. Cluster 5 showed strong specificity for Prostate Cancer, while Cluster 3 was enriched for Non-Small Cell Lung Cancer. Cluster 1, the group with the most favorable outcomes, was significantly enriched for Colorectal Cancer, suggesting the Colorectal signature in this cohort is substantially less aggressive than the Breast Cancer signature found in Cluster 2.

Cluster-Specific Drivers in Breast Cancer: To refine the genomic drivers, RSF analysis was restricted to the breast cancer cohort ($p < 0.0001$) (Figure 4). Cluster 4 (Blue) maintained the highest survival plateau, while Cluster 2 (Olive Green) dropped sharply, with survival probability falling below 25% significantly faster than all other cohorts. Global_cna_burden and TP53 mutations were identified as universal drivers across high-risk groups. However, cluster-specific signatures emerged where Cluster 5 was uniquely driven by GATA3 and TP53, while Cluster 3 relied on NF1. Tumor Mutational Burden (TMB_NONSYNONYMOUS) was highly predictive for Clusters 1 and 5 but had minimal impact on the stable Cluster 4. These results suggest that

distinct genomic paths, such as high CNA burden in Cluster 2 versus specific gene mutations in Cluster 5, can converge on different survival outcomes.

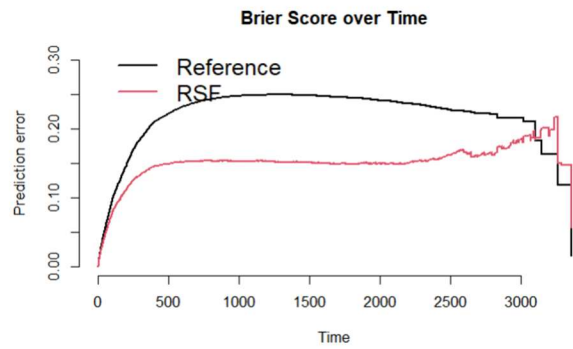


Figure 1: Time-Dependent Prediction Error (Brier Score) Comparison

Comparison of the prediction error (Brier score) over a follow-up horizon of approximately 3,361 days. The **RSF model (Red)** consistently demonstrates lower prediction error across the majority of the time-series compared to the **Reference model (Black)**

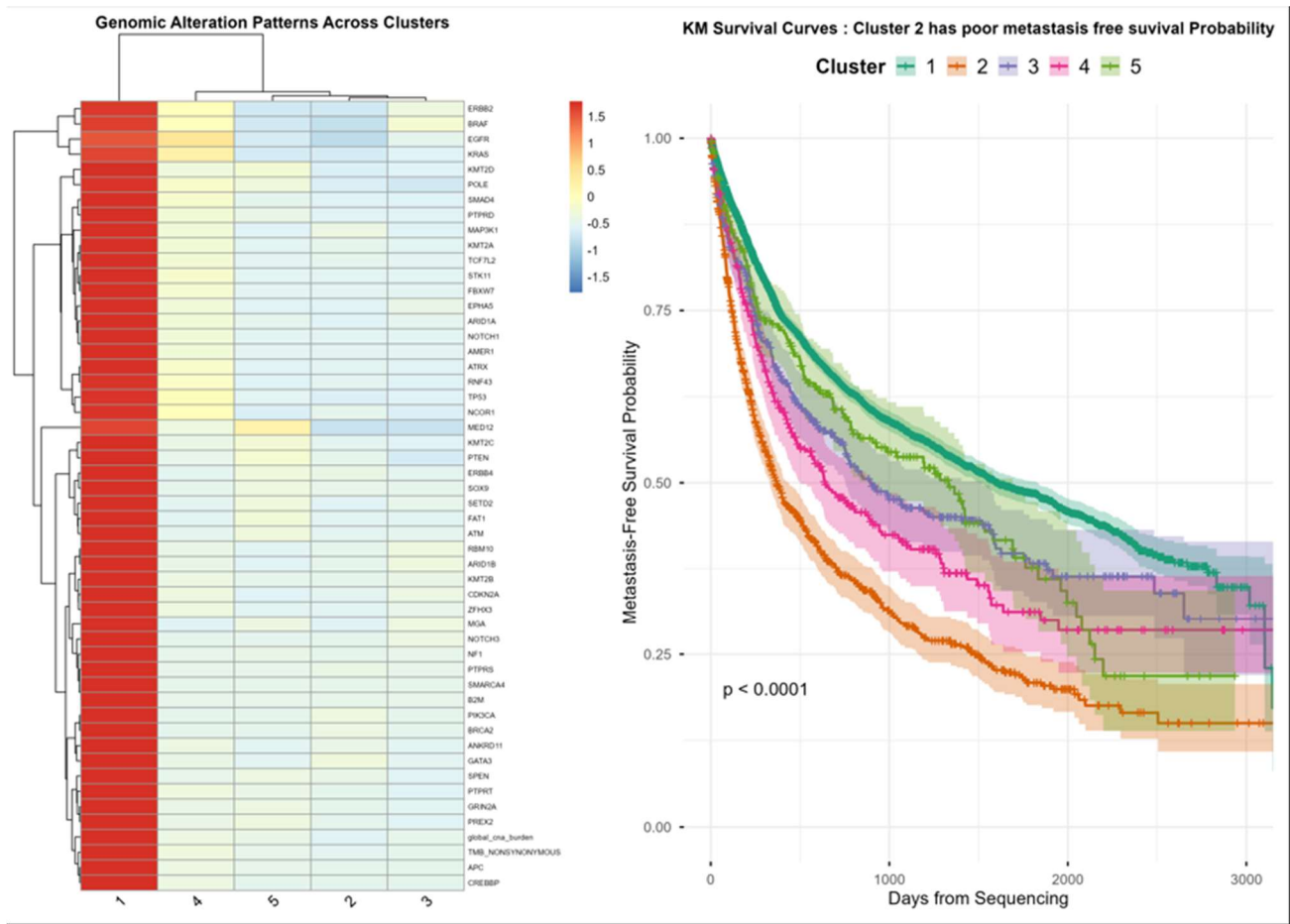


Figure 2: Integrated Genomic Landscape and Metastasis-Free Survival by Clusters

Left Panel: A hierarchical heatmap representing standardized genomic alteration patterns across five patient clusters identified through Random Survival Forest (RSF) proximity-based clustering. Rows represent specific genomic features (e.g., *TP53*, *ERBB2*, global CNA burden), while columns represent clusters. **Cluster 1** displays a distinct hyper-mutated phenotype across nearly all evaluated genes. **Right Panel:** Kaplan-Meier survival curves illustrating the probability of metastasis-free survival (MFS) over time (days from sequencing). Significant clinical divergence is observed ($p < 0.0001$), with **Cluster 2 (Orange)** exhibiting the most aggressive clinical trajectory and **Cluster 1 (Teal)** demonstrating a survival advantage despite its high mutational density. Shaded areas represent 95% confidence intervals.

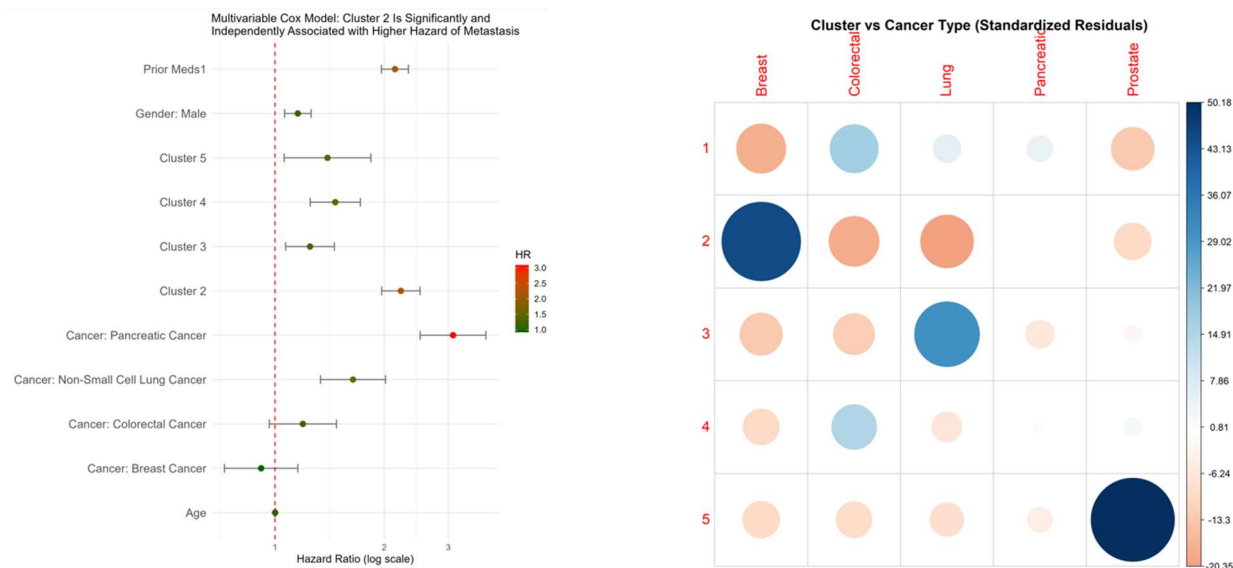


Figure 3: Multivariate Risk Assessment and Tissue-Specific Cluster Enrichment

Left Panel: A forest plot summarizing the results of a multivariate Cox proportional hazards model. Hazard Ratios (HR) are plotted on a log scale for genomic clusters and clinical covariates (Age, Gender, Prior Medication, and Cancer Type). Cluster 2 and Pancreatic Cancer emerge as the strongest independent predictors of metastatic risk, while Cluster 1 serves as the favorable reference group. **Right Panel:** A Pearson residual plot illustrating the association between molecular clusters (rows) and primary cancer types (columns). The size and color of the circles indicate standardized residuals; dark blue circles denote significant over-representation. Cluster 2 shows a dominant enrichment for Breast Cancer, while Cluster 5 is highly specific to Prostate Cancer, suggesting that while genomic signatures drive risk, they are strongly influenced by the tissue of origin

KM Survival curves and Top 20 Variable of Imp- Breast Cancer

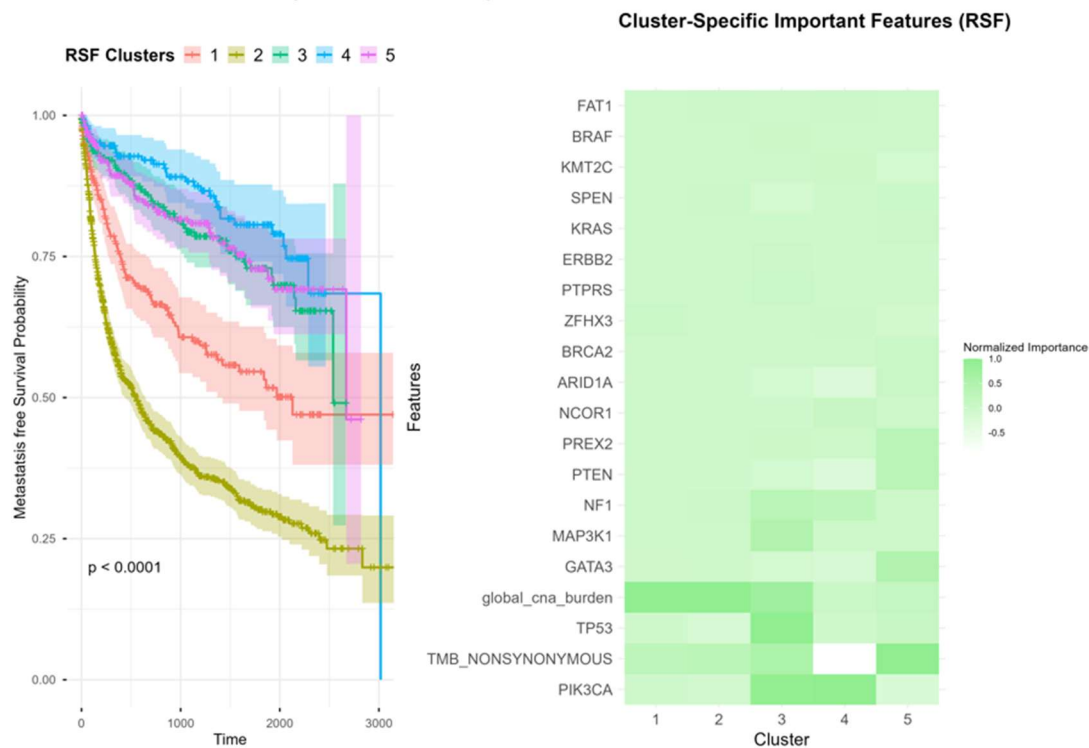


Figure 4: Survival Analysis and Feature Importance for Breast Cancer Stratification

Left Panel: Kaplan-Meier curves illustrating overall survival probability for five distinct patient clusters within the breast cancer cohort. A highly significant clinical divergence is observed ($p < 0.0001$), with Cluster 2 (Olive Green) representing the highest-risk group, characterized by a sharp decline in survival probability compared to the indolent trajectory of Cluster 4 (Blue). Shaded areas denote 95% confidence intervals. **Right Panel:** A heatmap of Cluster-Specific Important Features derived from the Random Survival Forest (RSF) model. The intensity of green indicates normalized variable importance (VIMP) for predicting survival within each subgroup. Universal drivers, such as global_cna_burden and TP53, show high importance across most clusters, while specific genes like GATA3 and NF1 exhibit cluster-specific predictive utility, highlighting the divergent Genomic paths influencing breast cancer mortality.

Conclusion & Discussion

This study successfully implemented an RSF clustering framework to move beyond traditional clinical staging and identify the genomic drivers of disease progression. By utilizing a Random Survival Forest proximity metric on a pan-cancer cohort of over 25,000 patients, five molecularly distinct subgroups were identified with highly divergent clinical trajectories. Kaplan–Meier analysis demonstrated significant differences in metastasis-free survival across the clusters (log-rank $p < 0.0001$), indicating that these genomic subtypes are not only genetically distinct but also clinically meaningful. The analysis of Cluster 1 revealed a counter-intuitive relationship between mutational density and clinical outcome. While the heatmap confirms Cluster 1 as a hyper-mutated, showing near-universal alterations across the gene panel, this group maintained the highest Metastasis Free Survival probability. This finding aligns with the neoantigen-load hypothesis⁹⁻¹²:

1. The elevated Tumor Mutational Burden (TMB) likely increases the production of neoantigens, rendering the tumor more susceptible to endogenous immune surveillance.
2. The extreme level of instability may impose a fitness penalty on the tumor cells, inhibiting their ability to successfully undergo the metastatic state

In contrast, **Cluster 2** emerged as a high-risk phenotype, particularly within breast cancer driven by heavy chromosomal instability and *TP53* mutations.

Ultimately, this research demonstrates that the integration of **supervised machine learning** with **unsupervised clustering** provides a powerful lens for precision oncology. This research demonstrates that unsupervised clustering on RSF-derived distances effectively reveals hidden, clinically relevant patient subtypes. This approach bridges the gap between genomic complexity and survival forecasting, offering a robust pipeline for precision medicine and the identification of high-risk patients who may benefit from intensified therapeutic interventions.

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